

MODULE 4

ALGEBRAIC STRUCTURE

INTRODUCTION

- we will study, binary operation as a function
- Algebraic structures- monoid, semigroups, groups and rings, integral domains, field.
- They are called an algebraic structure because the operations on the set define a structure on the elements of that set

BINARY OPERATION

Let A be non-empty set.

A function $f: A \times A \rightarrow A$ is called a binary operation on a set A

Generally the binary operation on A is denoted by $*$ then $a * b \in A \quad \forall a, b \in A$.

This property is described as **Closure Property** or A is closed under $*$

Examples

Q. Is $+$ binary operation on N (set of natural no)?

Ans : yes

Q. Is $+$ binary operation on Z , (set of integers)?

Ans : yes

Q. Is $-$ binary operation on N or Z^+ ?

Ans : No for N and Z^+

Q. Is $-$ binary operation on Z or R ?

Ans : yes

Q. Is a/b binary operation on Z , R , R^* ?

Ans : Only on R^*

Associative property

Let A be non empty set and $*$ is binary operation on A , then A is associative if $(a * b) * c = a * (b * c) \quad \forall a, b, c \in A$

Examples

Q. Is $+$ associative in $\mathbb{Z}$, $\mathbb{R}$?

Ans: yes

Q. Is $-$ associative in $\mathbb{Z}$, $\mathbb{R}$?

Ans: No

Q. Is multiplication/Division associative in $\mathbb{Z}$, $\mathbb{R}$?

Ans: multiplication yes/Division No

Commutative property

Let A be non empty set and $*$ is binary operation on A , then A is commutative if $(a * b) = (b * a) \quad \forall a, b \in A$

Examples

Q. Is usual addition(+)/usual multiplication($\times$) commutative in $\mathbb{Z}$, $\mathbb{R}$?

Ans: yes

Q. Is usual subtraction(-)/usual division(/) commutative in $\mathbb{Z}$, $\mathbb{R}$?

Ans: No

Identity Property

Let A be non empty set and $*$ is binary operation on A ,

If $e \in A$ and $a * e = a \forall a \in A$, then e is the identity element of A with respect to $*$

Examples

Q. What is the identity element of $\mathbb{R}$ with respect to addition?

Ans: 0

Q. What is the identity element of $\mathbb{R}$ with respect to multiplication?

Ans: 1

Inverse property

Let A be non empty set and $*$ is binary operation on A ,

If for $a \in A$ there exist $b \in A$ such that $a * b = e = b * a$ then 'b' is called an inverse of 'a' with respect to operation $*$

Example :

Q. What is the inverse of 3 in $\mathbb{R}$ with respect to addition?

Ans : -3

Q. What is the inverse of 3 in $\mathbb{R}^*$, $\mathbb{Z}$ with respect to multiplication?

Ans : $1/3$, Not available in $\mathbb{Z}$

SEMIGROUP

A non-empty set S together with a binary operation $*$ is called as a semigroup if binary operation $*$ is associative.

we denote the semigroup by $(S, *)$

Commutative Semigroup

A semigroup $(S, *)$ is said to be Commutative if $*$ is commutative

Examples

$(\mathbb{Z}, +)$ is a commutative semigroup

$(\mathbb{Z}, \cdot)$ is a commutative semigroup

Monoid

A non-empty set M together with a binary operation $*$ defined on it, is called as monoid if

- i) binary operation $*$ is associative
- ii) M has an identity with respect to $*$

Note: A semi group with an identity is a monoid
If $*$ is commutative, $(M, *)$ is called commutative monoid

Examples

$(\mathbb{Z}, +)$ is a monoid with identity 0

$(\mathbb{Z}, .)$ is a monoid with identity 1

$(\mathbb{N}, +)$ is a semigroup but not a monoid.

Group

A non-empty set G together with a binary operation $*$ defined on it, is called a group if

- (i) binary operation $*$ is closed
- (ii) binary operation $*$ is associative
- (iii) G has an identity with respect to $*$
- (iv) Every element in G has an inverse in G , with respect to $*$

We denote the group by $(G, *)$

Commutative (Abelian) Group : A group $(G, *)$ is said to be commutative if $*$ is commutative.

Examples

$(\mathbb{Z}, +)$ is an abelian group with identity 0 and $-a$ as inverse of a

$(\mathbb{R}, \cdot)$ is a monoid but not a group (identity 1, No inverse for 0)

$(\mathbb{R} - \{0\} / \mathbb{R}^*, \cdot)$ is an abelian group with identity 1 and $1/a$ as inverse of a .

$(\mathbb{Q} - \{0\} / \mathbb{Q}^*, \cdot)$ is an abelian group with identity 1 and $1/a$ as inverse of a .

Let $G = \{ M \mid M \text{ is } 2 \times 2 \text{ non-singular matrices} \}$
and $a * b$ is usual Matrix multiplication
then $(G, *)$ is Non-abelian group.

Example: Determine whether $S = \{1, 2, 3, 6, 12\}$ is a monoid, a semigroup or a group with respect to $*$ where $a * b = G.C.D.(a, b)$

$*$	1	2	3	6	12
1	1	1	1	1	1
2	1	2	1	2	2
3	1	1	3	3	3
6	1	2	3	6	6
12	1	2	3	6	12

Closure Property : Since all the elements of the table $\in S$, closure property is satisfied.

Associative Property : Since

$$a * (b * c) = a * (b * c) = a * GCD\{b, c\} = GCD\{a, b, c\}$$

$$\text{And } (a * b) * c = GCD\{a, b\} * c = GCD\{a, b, c\}$$

$$\therefore a * (b * c) = (a * b) * c$$

$\therefore *$ is associative.

$\therefore (S, *)$ is a semigroup.

Existence of identity: From the table we observe that $12 \in S$ is the identity

$\therefore (S, *)$ is a monoid.

Example: Determine whether $S = \{1, 2, 3, 6, 12\}$ is a monoid, a semigroup with respect to $*$ where $a * b = G.C.D.(a, b)$

Commutative: Since the table entries are symmetric, we will get $a * b = b * a$, Hence it is commutative.

$(S, *)$ is commutative monoid.

Check that No inverse for any element. Since in any row or column the identity 12 is not appearing.

Hence it is not a group .

Example : Determine whether $S = \{1, 2, 3, 6, 9, 18\}$ is a semigroup, a monoid or commutative monoid with respect to $*$ where $a * b = L.C.M.(a, b)$

$*$	1	2	3	6	9	18
1	1	2	3	6	9	18
2	2	2	6	6	18	18
3	3	6	3	6	9	18
6	6	6	6	6	18	18
9	9	18	9	18	9	18
18	18	18	18	18	18	18

Closure Property : Since all the elements of the table $\in S$, closure property is satisfied.

Associative Property : Since $a * (b * c) = a * LCM\{b, c\} = LCM\{a, b, c\}$

And $(a * b) * c = LCM\{a, b\} * c = LCM\{a, b, c\}$

$$\therefore a * (b * c) = (a * b) * c$$

$\therefore *$ is associative.

$\therefore (S, *)$ is a semigroup.

Example : Determine whether $S = \{1, 2, 3, 6, 9, 18\}$ is a semigroup, a monoid, commutative monoid with respect to $*$ where $a * b = L.C.M.(a, b)$

Existence of identity : From the table we observe that $1 \in S$ is the identity.

$\therefore (S, *)$ is a monoid.

Commutative property : Since $LCM\{a, b\} = LCM\{b, a\}$ we have $a * b = b * a$. Hence $*$ is commutative.

Therefore A is commutative monoid.

No inverse for any element. Hence it is not a group.

Example : Determine whether $A = \mathbb{Z} - \{1\}$, the set of integers except 1 is a semigroup or a monoid with respect to $*$ where $a * b = a + b - ab$

Closure Property : -

Let $a, b \in A = \mathbb{Z} - \{1\}$, the set of integers except 1

$\therefore a, b$ are integers and $a \neq 1, b \neq 1$

$\therefore a * b = a + b - ab$ is integer

Assume $a * b = 1 \Rightarrow a + b - ab = 1 \Rightarrow a + (1-a)b = 1$

$\Rightarrow 0 = 1 - a - (1-a)b \Rightarrow 0 = (1-a)(1-b)$

$\Rightarrow a = 1$ or $b = 1$ but given $a \neq 1, b \neq 1$

$\therefore$ Assumption $a * b = 1$ is wrong $\Rightarrow a * b \neq 1$

$a * b = a + b - ab$ is integer and $a * b \neq 1 \Rightarrow a * b \in A = \mathbb{Z} - \{1\}$

$\therefore a * b \in A \quad \forall a, b \in A.$

so $*$ is binary operation (Or $*$ satisfies closure property)

Associative Property:

$$\begin{aligned} G1 : a * (b * c) &= a * (b + c - bc) = a + (b + c - bc) - a(b + c + bc) \\ &= a + b + c - bc - ab - ac - abc \end{aligned}$$

$$\begin{aligned} \text{And } (a * b) * c &= (a + b - ab) * c = (a + b - ab) + c - (a + b + ab)c \\ &= a + b + c - ab - ac - bc - abc. \end{aligned}$$

Hence, $a * (b * c) = (a * b) * c$. $\therefore *$ is associative.

Hence $(A, *)$ is Semi-Group

Existence of identity:

Let e be the identity element

$$a * e = a = e * a$$

$$a * e = a + e - ae = a$$

$$e(1-a) = 0$$

Either $e = 0$ or $a = 1$

But $a \neq 1$

$e = 0$ is the identity element

(Check for commutativity !!)

Hence $(A, *)$ is commutative monoid

Check it is Not group (No inverse)

What if $A = R - \{1\}$?

Example: Prove that A is a group with respect to $*$

Where $A = \mathbb{R} - \{1\}$ the set of real numbers except 1

And $a * b = a + b - ab$

Closure, Associative, commutative and identity element:

Same arguments like last example, just replace integers ($\mathbb{Z}$) by Real No ($\mathbb{R}$)

Existence of Inverse:

Let b be the inverse of a then $a * b = e = b * a$

$$a + b - ab = 0$$

$$a + b(1-a) = 0$$

$b = -a/(1-a)$ and $-a/(1-a)$ is real number as $a \neq 1$

Hence Inverse of a with respect to $*$ is $-a/(1-a) \in A$.

A is an abelian group with respect to $*$

(operation for infinite, Cayley table for upto 10 elements)

Results

If G is a group.

- (i) Then its identity element is unique.
- (ii) each a in G has unique inverse

Example : Consider $G = \mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$. Then Prepare table for addition modulo n in G . Hence find identity element and inverse of 2,3,6. Is G group under addition modulo n ?

	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	0
2	2	3	4	5	6	0	1
3	3	4	5	6	0	1	2
4	4	5	6	0	1	2	3
5	5	6	0	1	2	3	4
6	6	0	1	2	3	4	5

Now Check for properties !!

Identity is 0 and inverse of a is $7 - a$

Example : Prepare table for multiplication modulo n for $G = \mathbb{Z}_7 - \{0\}$ Hence find identity element and inverse of 2,3,6

*	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

From the table we observe that $1 \in G$ is identity.

From the table we get $2^{-1} = 4$, $3^{-1} = 5$, $6^{-1} = 6$

Ring

$(R, \oplus, \otimes)$ is said to be ring if

- (i) $(R, \oplus)$ is a commutative group
- (ii) $(R, \otimes)$ is a semigroup
- (iii) $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$ ($\otimes$ distributes over $\oplus$)

Field

$(R, \oplus, \otimes)$ is said to be field if

- (i) $(R, \oplus)$ is a commutative group
- (ii) $(R - \{0\}, \otimes)$ is a commutative group, where 0 is identity w.r.t. $\oplus$
- (iii) $a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c)$ ($\otimes$ distributes over $\oplus$)

Example : Prove that $(\mathbb{Z}_5, +, \cdot)$ is field

Answer: To prove $(\mathbb{Z}_5, +)$ & $(\mathbb{Z}_5 - \{0\}, \cdot)$ are commutative groups

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

x	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

Example: Prove that set $\{\bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}\}$ is a commutative ring modulo 10.

+	0	2	4	6	8
0	0	2	4	6	8
2	2	4	6	8	0
4	4	6	8	0	2
6	6	8	0	2	4
8	8	0	2	4	6

×	0	2	4	6	8
0	0	0	0	0	0
2	0	4	8	2	6
4	0	8	6	4	8
6	0	2	4	6	8
8	0	6	2	8	4

We can prove that this is a field also.

Commutative Ring

$(R, \oplus, \otimes)$ is said to be commutative ring if

- (i) $(R, \oplus, \otimes)$ is a ring
- (ii) $\otimes$ is commutative

Ring with unity

$(R, \oplus, \otimes)$ is said to be ring with unity if

- (i) $(R, \oplus, \otimes)$ is a ring
- (ii) Identity w.r.t. $\otimes$ exists in R

Definition: Integral Domain

$(R, \oplus, \otimes)$ is said to be Integral Domain if

- (i) $(R, \oplus, \otimes)$ is commutative ring with unity
- (ii) R has no zero divisors

Zero divisors

$(R, \oplus, \otimes)$ is ring

if $a \otimes b = 0$ (0 is identity w.r.t. $\oplus$)

but $a \neq 0$ & $b \neq 0$ then a & b are said to be zero divisors

Example

Find zero divisors in ring $(\mathbb{Z}_6, +, \cdot)$

$2 \cdot 3 = 0$ but $2 \neq 0, 3 \neq 0$

$4 \cdot 3 = 0$ but $4 \neq 0, 3 \neq 0$

2 & 3, 4 & 3 are zero divisors of ring $\mathbb{Z}_6$

Definition: Units

$(R, \oplus, \otimes)$ is ring and 1 is identity w.r.t. $\otimes$
if $b \in R$ is inverse of 'a' w.r.t. $\otimes$ then a & b are called units of ring R

Example :

Find units in ring $(Z_9, +, \cdot)$

$$2 \cdot 5 = 1$$

Then 2 & 5 are units of Z_9

Find other units of Z_9 .

Definition: Integral Domain

$(R, \oplus, \otimes)$ is said to be Integral Domain if

- (i) $(R, \oplus, \otimes)$ is commutative ring with unity
- (ii) R has no zero divisors

Example : Prove that Ring $(\mathbb{Z}_5, +, \cdot)$ is Integral Domain
Is Ring $(\mathbb{Z}_6, +, \cdot)$ Integral Domain ?

Note :

Ring $(\mathbb{Z}_p, +, \cdot)$ is Integral Domain and field if p is prime

In $\mathbb{Z}_n$, a is unit if $\text{G.C.D}(a, n) = 1$

In $\mathbb{Z}_n$, a is zero divisor if $\text{G.C.D}(a, n) \neq 1$